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# Multi-Page Visually Rich Document Understanding

A survey of system design and an empirical attribution of failure

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Lewei Xu

Supervised by Dr. Yihao Ding · Dr. Daochang Liu

# Roadmap

## 01 Background

Multi-page, visually rich document understanding: what the documents are, what the task is, and how the field got here.

## 02 Survey

*Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding*

EMNLP 2026, main conference

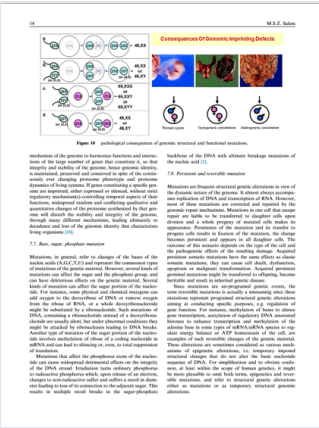
## 03 Empirical attribution study

*Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution*

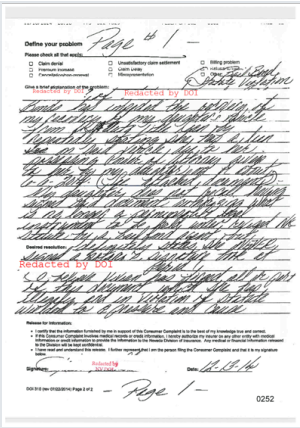
Submitted to EACL, under review

# Multi-Page Visually Rich Documents

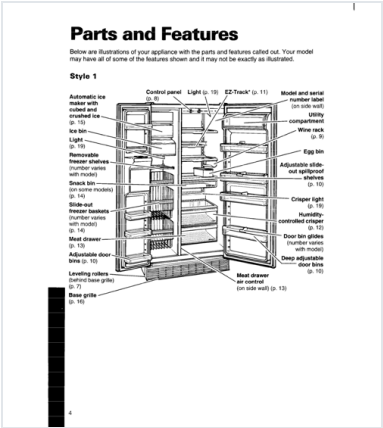
Meaning is carried jointly by **text, vision, and layout** rather than by text alone. **Multi-page** adds a further layer: meaning also runs between pages, as when a claim points to a figure many pages away.



Research paper  
born-digital



Signed form  
scanned, handwritten



Technical manual  
layout-dependent

Table 1: Consolidated Statement of Operations

|                                     | 2004         | 2003         | 2002         | 2001         |
|-------------------------------------|--------------|--------------|--------------|--------------|
| Revenue                             | \$ 1,770,101 | \$ 1,588,474 | \$ 1,575,762 | \$ 1,889,342 |
| Operating income                    | \$ 104,400   | \$ 104,400   | \$ 104,400   | \$ 104,400   |
| Income before taxes                 | \$ 104,400   | \$ 104,400   | \$ 104,400   | \$ 104,400   |
| Income tax expense                  | \$ 25,100    | \$ 25,100    | \$ 25,100    | \$ 25,100    |
| Net income                          | \$ 79,300    | \$ 79,300    | \$ 79,300    | \$ 79,300    |
| Weighted average shares outstanding | 422,000      | 422,000      | 422,000      | 422,000      |
| Earnings per share                  | \$ 0.19      | \$ 0.19      | \$ 0.19      | \$ 0.19      |

Table 2: Consolidated Statement of Cash Flows

|                                       | 2004         | 2003         | 2002         | 2001         |
|---------------------------------------|--------------|--------------|--------------|--------------|
| Net cash used in operating activities | \$ (146,000) | \$ (146,000) | \$ (146,000) | \$ (146,000) |
| Net cash used in investing activities | \$ (146,000) | \$ (146,000) | \$ (146,000) | \$ (146,000) |
| Net cash used in financing activities | \$ (146,000) | \$ (146,000) | \$ (146,000) | \$ (146,000) |

Financial report  
dense tables



Presentation deck  
text in images

- **Text.** Small, typeset or handwritten, and legible only at far higher input resolution than a typical vision-language task requires.
- **Vision.** Charts, figures, photographs, and handwriting carry meaning that no text extraction recovers.
- **Layout.** Position is semantic, and reading order must be recovered rather than assumed.

# What is Document Understanding?

One broad way to define it: get a system to **answer a natural-language question over a document**, accurately. The systems range from rule-based lookup, through trained vision-language models, to pipelines built around pretrained general-domain ones.

- 1

"What year was this report published?"

FACT LOOKUP
- 2

"According to the table, which region had the highest revenue in 2023?"

READ A TABLE
- 3

"How much did the leading region's revenue grow between the 2022 and 2023 results tables?"

TWO TABLES, THEN ARITHMETIC
- 4

"Does the growth described in the executive summary agree with the figures in the results table?"

PROSE AGAINST NUMBERS
- 5

"Across the report and its appendix, what evidence supports the conclusion that the new method improved performance, and what limitation qualifies it?"

GATHER, THEN QUALIFY

Illustrative questions, not quoted benchmark items

# A Brief History

A useful starting point is **MP-DocVQA** in 2023: a benchmark of documents up to twenty pages, with questions answerable from a single page, plus the Hi-VT5 baseline. It is the pivotal moment at which long-document understanding became its own problem.

## BENCHMARKS

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### MP-DocVQA, 2023

Up to twenty pages. One page carries the answer, so the task is finding it.

### MMLongBench-Doc, LongDocURL

- Documents into the hundreds of pages, with questions that combine evidence from several places at once.

## MODELS

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### Purpose-built, trained end to end

Hi-VT5 and its successors: a document encoder trained for this setting specifically.

### General-domain VLMs as the backbone

- Increasingly sophisticated training and training-free machinery built around a pretrained model that already reads text and page images well.

A lot happened in between, and quickly. Organising that is what the survey does.

# Document Questions in Consumer Assistants

Everyone here has used these. What changed is not the question being asked, but **how much of the document the model has to hold at once**.

A YEAR OR TWO AGO

## The whole document, in context

Parsed text plus an image of every page, all placed in the model's context window. Fine for short documents, and it hits the ceiling fast.

SCREENSHOT  
"cannot upload document,  
out of context" error

MORE RECENTLY

## On disk, read by tool calls

The upload lands in the model's working environment. It reads what it needs, when it needs it, rather than ingesting everything up front.

SCREENSHOT  
assistant reading an uploaded  
file through a tool call

AND NOW

## Projects, searched by retrieval

Upload many documents at once. Beyond what fits in context, the content is indexed and retrieved against the question instead.

SCREENSHOT  
a project with several  
uploaded documents

# Real-World Constraints

A hospital, a mining company, or a law firm wants the same thing: **classification, summarisation, or structured extraction over tens of thousands of documents**. The consumer path is often not available to them.

## WHY NOT JUST CALL AN API

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- **The documents cannot leave.** Medical records, legal and financial filings, and resources data carry privacy and data-residency requirements that rule out a third party.
- **The volume makes it uneconomic.** At tens of thousands of documents, per-call pricing stops being worth the benefit even where sending the data is permitted.
- **The document type is unusual.** A niche form or report may need its own training or fine-tuning rather than whatever a general model already knows.

## WHAT THAT FORCES

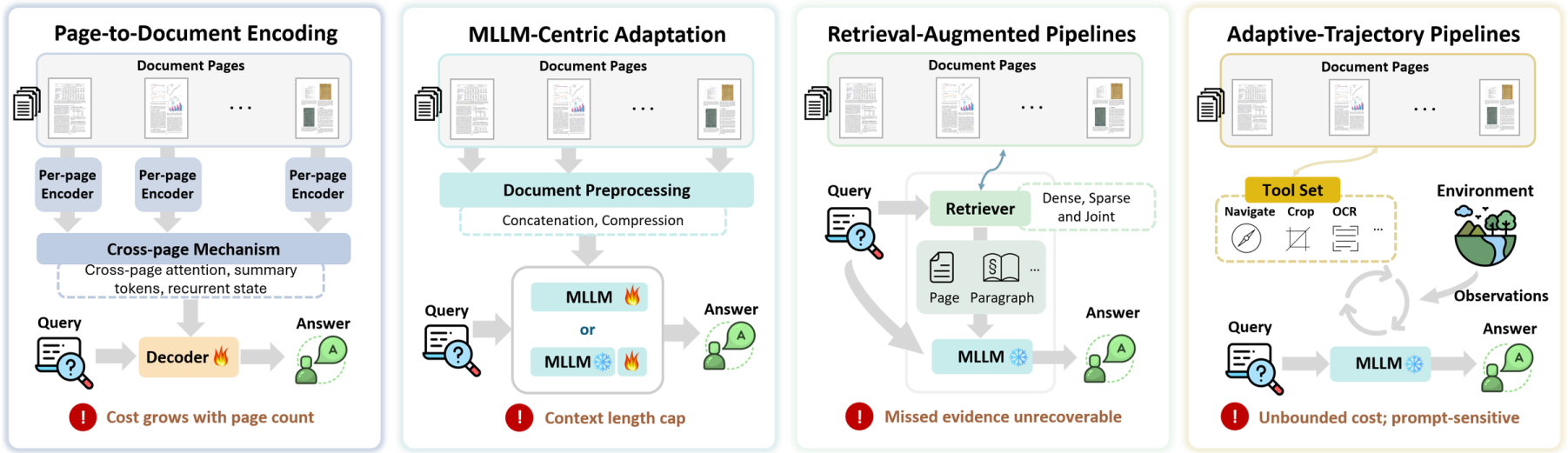
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- **Open-weight models, run locally.** A pretrained vision-language model on consumer hardware, not a datacentre.
- **A fixed compute budget.** Which pages are read, at what resolution, and how many times, all have to fit inside it.
- **Efficiency as a research problem.** Cost and latency are open questions in their own right, not implementation details.

This is where the research sits: **techniques and architectures for document understanding under a fixed compute budget**. The same techniques then flow back into the consumer products, which is why those have moved so quickly.

# Multi-Page Architectural Overview

The survey groups systems by **how they manage evidence across pages**. One family encodes the pages together; one adapts a multimodal model to take more pages directly; one retrieves relevant pages before generating; one inspects pages step by step, deciding what to look at next.



SYSTEMS PER YEAR, FROM THE SURVEY CORPUS · 46 SYSTEMS, 2023 ONWARD

|                           | 2023 | 2024 | 2025 | 2026 | total |
|---------------------------|------|------|------|------|-------|
| Page-to-document encoding | 1    | 2    | 1    | .    | 4     |
| MLLM-centric adaptation   | .    | 2    | 4    | 5    | 11    |
| Retrieval-augmented       | .    | 4    | 11   | 5    | 20    |
| Adaptive-trajectory       | .    | .    | 6    | 5    | 11    |

Roughly chronological: **page-to-document opens the field and thins out**, retrieval takes over in 2025, and adaptive-trajectory systems appear only in 2025.

# Text Modality

Text is the cheap modality: we can **search it and read it efficiently**. The problem is that a long document contains far too much of it.

- **Volume.** A system either compresses every page into a fixed budget, or selects only the passages it thinks it needs. Both are lossy, and either can discard the thing that turns out to matter.
- **Continuity.** Paragraphs, tables, and section context run across page boundaries, so a chunk cut at the wrong place loses the context that makes it readable.
- **Noise.** OCR and parsing errors accumulate across pages. Because extracted text is usually also the retrieval index, those errors degrade finding as well as reading.
- **Heterogeneity.** Text density varies sharply from page to page, yet almost every system applies one fixed budget to the whole document.

Some evidence cannot be recovered from text at all. **That is where the page image matters.**

# Visual Modality

Page images preserve what a text extract misses: **charts, handwriting, fine layout, and visual relationships**. What they cost is tokens.

- **Resolution.** Token cost scales with page count multiplied by per-page resolution, and fine-grained content is only legible at the high end of that range.
- **So: compress, or select.** Keep every page in context at reduced fidelity, or read a retrieved subset at full fidelity. The first blurs small text and chart annotations; the second never sees a page the retriever missed.
- **Signal competition.** Within one page, the same visual budget has to carry text-bearing pixels, figures and charts, and layout cues at once.

The trade is always the same: **broad page coverage against readable detail**. Beyond text and images, a document also has relationships that connect its pages.

# Cross-Page Structure Modality

A long document is **more than a stack of independent pages**. Its structure is what turns isolated page observations into document evidence.

- **Hierarchy and reading order.** Sections, lists, and captions tell us how the parts fit together, and the same evidence can mean different things depending on which section it sits under.
- **Cross-page references.** A sentence on one page points to a figure on another. Resolving “see Figure 3” to its referent is not something similarity scoring can do.
- **Spanning elements.** Tables and arguments continue across page boundaries, so the useful unit is larger than a page but smaller than the document.

Systems recover this by building outlines, trees, or graphs over the document. **Errors in that structure then mislead every step that follows.**

# Retrieval and Navigation

When a question arrives, retrieval **narrows the document to a smaller set of pages or passages**, so the reasoner never has to process the whole thing.

- **Four ways to search.** By words, with a sparse lexical scorer. By learned similarity over text embeddings. By image, with late-interaction encoders that score the question against the rendered page directly. Or by structure, walking a tree or graph built over the document.
- **Structure is what similarity cannot do.** Scoring passages independently handles paraphrase well, but it cannot follow a cross-reference or a dependency that no single passage carries.
- **The main risk is omission.** If a page the answer requires never reaches the reasoner, nothing downstream can use what was on it.

Finding the evidence is only half the task. **The model still has to reason with it.**

# Reasoning Strategies

Once the evidence is assembled, the model still has to make sense of it. Some answers need **facts from different pages combined, step by step**.

- **Ask it to show its reasoning.** Chain-of-thought elicits the intermediate steps before the answer, which binds evidence from different pages one hop at a time and leaves a trace you can check.
- **Ask it to check itself.** Self-reflection has the model audit or revise an initial answer against the same evidence, optionally with a separate verifier.
- **Ask it several times.** Sampling multiple reasoning paths and taking the majority buys robustness against a single bad chain, and costs proportionally more compute.

All of it operates over a **fixed buffer of evidence assembled before the call**. None of it can recover evidence that was never supplied.

# Agentic Methods

An agentic system **decides what to inspect next as it goes**, rather than executing a plan fixed in advance. Each observation feeds the next decision.

- **It can go back for more.** Search again with a refined query, follow a cross-reference to another section, or request a closer look at a page it has already seen.
- **It can divide the work.** Some systems split the roles: a reader that gathers, a planner that decides, a checker that decides whether the answer is good enough.
- **Compute follows difficulty.** An easy question stops early; a hard one keeps going. The trajectory also becomes an audit trail of what the system actually looked at.

The trade-off is **time, and more opportunities for a bad decision to lead the system astray**. It also still inherits the recall ceiling of every tool it calls.

# Training Strategies

Almost every system starts from a model **already trained on text and images**. What gets trained after that is the document-specific machinery around it.

## Find

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A retriever learns to surface better pages: aligning visual page embeddings with parsed text, folding in neighbouring-page context, or scoring pages by their role in a reasoning path rather than by similarity alone.

RETRIEVER

## Combine

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A reasoner learns to produce structured reasoning on its own, by imitating demonstration traces and then by reinforcement against rewards that credit evidence recall as well as the final answer.

REASONER

## Decide

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A controller learns when to inspect more and when to stop, usually by distilling a stronger model's trajectories first and then optimising the policy directly.

CONTROLLER

Each of those improvements depends on having suitable training examples. **That makes the available datasets a central part of the field.**

# Datasets

Datasets serve three roles, and the multi-page setting makes all three harder: **per-page annotation does not scale** to documents of tens or hundreds of pages, and long PDFs are harder to source and license than single pages.

## Pretraining

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The exception rather than the norm. Genuinely multi-page corpora have appeared only recently, so most systems inherit a backbone pretrained on single pages and skip the stage entirely.

RARE

## Fine-tuning

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Where the weight falls. Broad instruction mixes of released question-answer collections, plus bespoke corpora synthesised by a strong model over scraped PDFs.

MOSTLY SYNTHETIC

## Benchmarking

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Accumulated in tiers, from single-page extraction through to long documents whose evidence spans pages. In practice evaluation has consolidated on a small, much-reused pool.

NARROW POOL

Benchmarks measure whether the answer was right. For this problem we also want to know **whether the right evidence was found and used** to get there.

# Why Locate the Failure?

The survey shows many ways to build these systems. But when a system answers wrongly, **different papers give different explanations for why**, and those explanations imply different fixes.

“How much did the leading region's revenue grow between the 2022 and 2023 tables?”

The system answers **14%**. The correct answer is 23%. What went wrong?

## The page was read badly

The parser garbled a figure in the table, or the page was rendered too small to read it.

## The right page was missed

Only one of the two tables was retrieved, so the comparison was never possible.

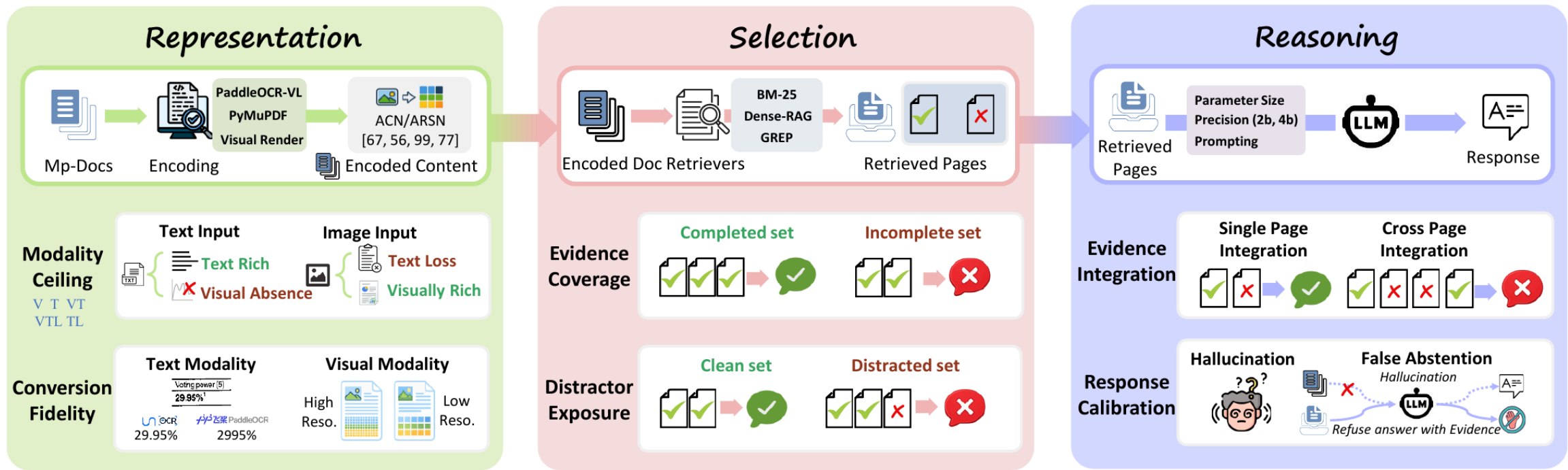
## The model could not use it

Both tables were present and legible, and it still picked the wrong row or did the arithmetic wrong.

End-to-end accuracy cannot tell these apart, and **under a fixed compute budget you have to know which one to spend on**.

# Three Failure Loci

Three places in the pipeline where an answer can go wrong, each with its own diagnostic question.



## REPRESENTATION

Did we preserve the information the question needs from the page?

## SELECTION

Did that information actually reach the model?

## REASONING

Did the model use it correctly once it had it?

The order matters, and it is what makes attribution possible: **we cannot reason over evidence that was lost in encoding, or never selected.** So a locus can only be tested once the ones above it are neutralised.

# How We Isolate a Failure

One simple pipeline throughout, with **one condition changed at a time** and everything else held fixed.

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## Encode

Every page becomes text, an image, or both.  
Done once per document and cached, so no question changes what a page contains.

PaddleOCR-VL · 200 DPI

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## Retrieve

▶ A retriever scores every encoded page against the question and returns the top  $k$ , restored to document order.

ColQwen3 · page-level

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## Answer

▶ A multimodal model receives those pages and the question, and answers in a single pass. Nothing is fed back.

Qwen3-VL-8B · greedy

- **The stages act once, and in order.** That is what lets a failure be attributed rather than guessed at.
- **Gold pages bypass retrieval.** For most tests the annotated evidence pages are supplied directly, so selection is neutralised and whatever remains is representation or reasoning.
- **Every reported accuracy carries a confidence interval** from a document-level bootstrap, because questions from one document are not independent.

# Representation Experiments

Two questions about the page itself: **what can an input express**, and **how much is lost converting the page into it**.

## Modality ceiling

WHAT THE INPUT CAN EXPRESS

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Four rungs of increasing richness. **T** is the PDF's embedded text layer. **TL** is layout-aware parser markdown. **TLV** adds the rendered page image. **V** is the page image alone.

The comparison asks whether the required evidence is still recoverable once the textual, structural, or visual signal is taken away, rather than assuming one representation is best everywhere.

## Conversion fidelity

HOW MUCH SURVIVES THE CONVERSION

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Hold the rung fixed and vary the quality of each channel instead: **the parser** for text, across PyMuPDF, MinerU, PaddleOCR-VL and Unlimited-OCR, and **the render resolution** for the image.

This separates what a modality cannot carry in principle from what a bad conversion destroyed in practice.

The annotated evidence pages are supplied in both, so **retrieval is not a variable here**. Whatever changes is the encoding.

# Selection Experiments

Selection can fail in two opposite ways, and the point of the experiment is that **they are not equally damaging**. Both start from the pages known to contain the answer.

## Evidence coverage

REQUIRED EVIDENCE REMOVED

Take questions that cite more than one gold page and **withhold one of them**. Pages are ranked within the document, so we can remove the highest-ranked gold page, the lowest-ranked, or keep only one.

A question that cannot satisfy a condition is excluded and recorded, never quietly satisfied with a different page set.

## Distractor exposure

IRRELEVANT EVIDENCE ADDED

Keep the complete gold set and **add up to five top-ranked non-gold pages** alongside it. Conditions are blocked on the exact gold-page count, so questions needing different amounts of evidence are never pooled.

The supplied context stays at six pages or fewer, so distractor count is not confounded with context length.

The encoding and the reasoner stay fixed across both, so **the two effects can be compared directly** against the same oracle baseline.

# Reasoning Experiments

With representation and selection controlled, what remains is the reasoner. Two different problems, so **they are tested separately**.

## Evidence integration

CAN IT COMBINE WHAT IT HAS?

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Split the questions by the **structure of the evidence** they need: one page and one source, several sources on one page, or evidence distributed across pages.

Then add a step-by-step reasoning prompt and see which of those groups it actually helps. The annotated pages are supplied throughout, so any deficit arises after the evidence has arrived.

## Response calibration

SHOULD IT ANSWER AT ALL?

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Vary both whether the evidence supports an answer and whether the model is **explicitly permitted to abstain**.

Answerable questions get their oracle gold pages. Unanswerable ones get the three top-ranked pages by BM25: plausible context that does not support an answer. That measures unsupported answers in one direction and false refusals in the other.

Everything is decoded greedily, so a condition is **reproducible from its identity** rather than from a seed.

# Datasets Make Attribution Possible

None of these experiments are possible without annotation that says **which pages support each question**. That is what picks the benchmarks.

## MMLongBench-Doc

PRIMARY DIAGNOSTIC

|           |            |            |              |
|-----------|------------|------------|--------------|
| 135       | 47.5       | 847        | 244          |
| documents | mean pages | answerable | unanswerable |

The only multi-page benchmark we know of that annotates, per question, **the evidence pages, the kind of evidence it draws on, and the document's domain**, while also supplying plausible questions the document cannot answer.

Evidence pages make coverage measurable and define the oracle. Evidence source explains representation results by what the question actually needs. The unanswerable pool makes calibration testable rather than inferred.

Answers are scored by an LLM judge rather than exact match, since a correct response can differ in units, rounding, or formatting. The judge sees only the question, gold answer, and model answer, never the condition being tested. It agrees with GPT-4o on **96.0%** of a stratified sample and with a human annotator on **98.3%**. We track refusals and wrong answers separately, not just accuracy.

## LongDocURL

SECONDARY REPLICATION

|           |            |           |              |
|-----------|------------|-----------|--------------|
| 396       | 85.6       | 2,325     | 8            |
| documents | mean pages | questions | unanswerable |

Longer documents, used to check whether the findings hold under a **less demanding setting**. Born-digital throughout, almost no unanswerable questions, and many questions answerable from the text even when the annotated evidence is visual.

Its annotated pages are also not a minimal set: several may be redundant or independently sufficient, which we account for when reading its coverage and integration results.

# Text and Vision Complement Each Other

With the correct pages supplied, accuracy climbs the representation ladder: **29.0% text only**, **42.6% image only**, **49.5% with both**.

| Evidence source | Accuracy    |             |             |             | Abstention  |            |            |            | Incorrect   |             |             |             |
|-----------------|-------------|-------------|-------------|-------------|-------------|------------|------------|------------|-------------|-------------|-------------|-------------|
|                 | T           | TL          | TLV         | V           | T           | TL         | TLV        | V          | T           | TL          | TLV         | V           |
| Text            | <b>36.9</b> | 44.4        | <b>51.0</b> | 42.3        | 23.0        | 11.5       | 7.3        | 8.0        | 40.1        | 44.1        | 41.6        | 49.7        |
| Layout          | 26.3        | 33.9        | 49.2        | <b>43.2</b> | 20.3        | 10.2       | <b>0.8</b> | <b>4.2</b> | <b>53.4</b> | <b>55.9</b> | <b>50.0</b> | 52.5        |
| Table           | 35.6        | <b>44.7</b> | 44.7        | 34.1        | <b>15.9</b> | <b>7.2</b> | 6.7        | 9.1        | 48.6        | 48.1        | 48.6        | <b>56.7</b> |
| Chart           | 16.3        | 16.9        | 42.7        | 40.4        | 32.6        | 28.1       | 7.3        | 6.2        | 51.1        | 55.1        | <b>50.0</b> | 53.4        |
| Figure          | 14.5        | 22.1        | 42.4        | 39.0        | 36.6        | 24.8       | 7.6        | 7.9        | 49.0        | 53.1        | <b>50.0</b> | 53.1        |
| All             | 29.0        | 35.7        | 49.5        | 42.6        | 26.3        | 17.2       | 6.2        | 7.1        | 44.7        | 47.2        | 44.3        | 50.3        |

Answer outcomes by evidence source and representation rung, MMLongBench-Doc, oracle pages

## – Text alone misses visual questions

At **T**, chart questions reach 16.3% and figure questions 14.5%. Adding the page image takes them to 42.7% and 42.4%.

## – It fails by refusing, not by guessing

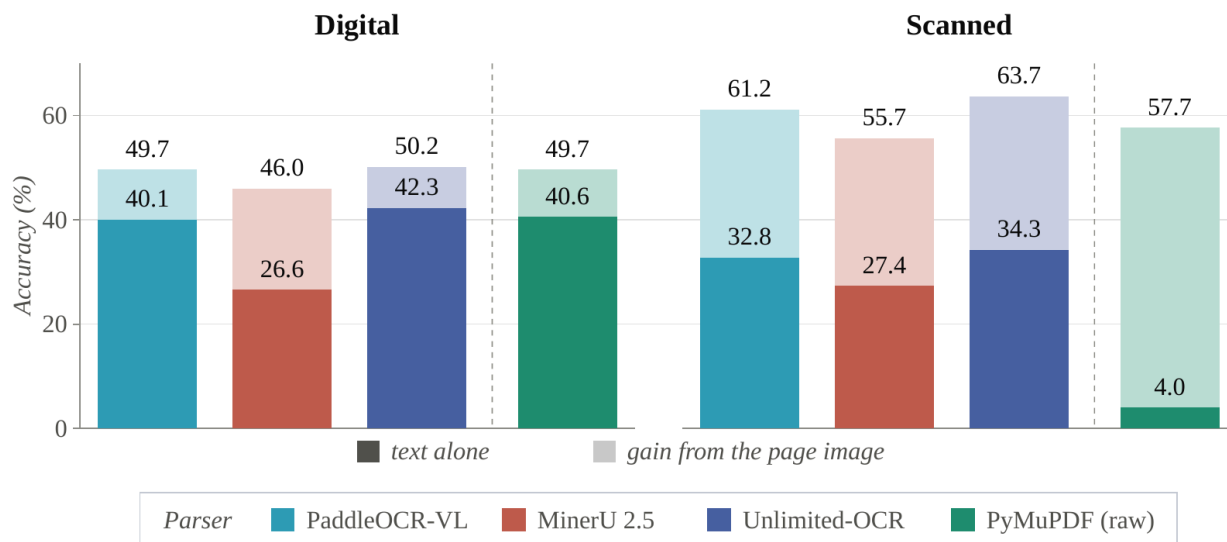
Abstention on figure questions falls from 36.6% to 7.6% across the ladder, while the incorrect rate stays broadly flat. The gain comes out of refusals.

## – Images alone are weaker than both

**V** reaches 42.6% against 49.5% for **TLV**. Text exposes fine detail in a form that is easier to use than recovering it from pixels.

# Conversion Quality Matters

A useful input type still fails if the page is converted badly. **Parser choice alone moves accuracy by tens of points** — it is not neutral preprocessing.



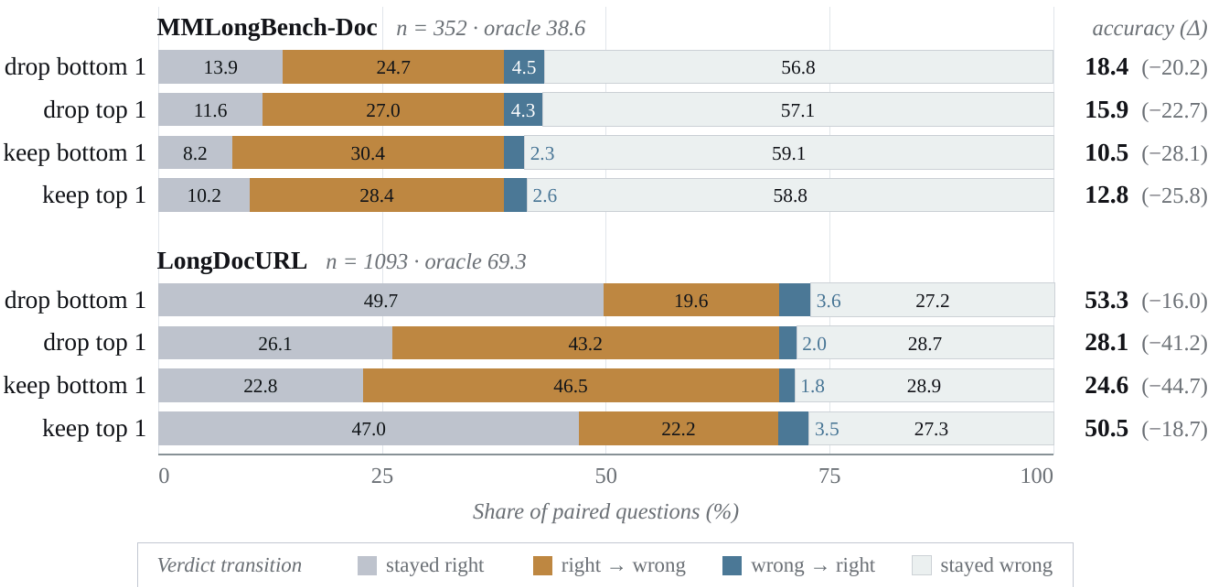
Parser accuracy by text source, with and without the page image

## READING IT

- **Scans break the text channel entirely.** On scanned documents the PDF's embedded text layer is often absent, and accuracy collapses. Conversion failure does not merely degrade the representation, it removes it.
- **Parsers differ a lot.** MinerU is the weakest on both digital and scanned pages; PaddleOCR-VL and Unlimited-OCR recover substantially more.
- **Vision covers for the parser.** Adding the page image compresses the spread between parsers sharply, and the weakest parser benefits most. Many conversion errors are recoverable once a second channel still holds the signal.
- **Resolution is the mirror image.** Raising render resolution helps **V** far more than **TLV**, where the text already carries part of what the extra pixels would recover.

# Missing Evidence Is Costly

Now remove one page the question needs. **Answers that were correct become wrong, and the reverse almost never happens.**



Verdict transitions after withholding or retaining gold evidence pages

## READING IT

- Withholding one gold page breaks **24.7–27.0%** of previously correct answers, while only 4.3–4.5% are repaired. Keeping just one gold page widens it further: 28.4–30.4% break against 2.3–2.6% repaired.
- The loss is directional. This is more than the expected correlation between retrieval recall and accuracy. Removing known evidence produces a one-way loss that downstream reasoning rarely reverses.
- Which page survives matters too. Keeping the highest-ranked gold page preserves much more than keeping the lowest. Extractive questions tolerate partial evidence; counting and calculation do not.

# Moderate Distraction Is Less Damaging

The opposite error: retrieval returns the right pages **plus a few extra**. Within the range we test, this costs far less than missing a required page.

| Condition           | MMLongBench-Doc |      |      |      |      |      | LongDocURL |      |      |      |      |     |
|---------------------|-----------------|------|------|------|------|------|------------|------|------|------|------|-----|
|                     | T               | TL   | TLV  | V    | R→W  | W→R  | T          | TL   | TLV  | V    | R→W  | W→R |
| <i>1 gold page</i>  |                 |      |      |      |      |      |            |      |      |      |      |     |
| Oracle              | 37.1            | 44.6 | 63.7 | 55.0 | —    | —    | 65.6       | 74.0 | 74.7 | 64.5 | —    | —   |
| +1 distractor       | 39.5            | 48.9 | 63.7 | 55.7 | 7.8  | 7.0  | 65.4       | 72.4 | 74.4 | 63.0 | 8.5  | 8.1 |
| +2 distractors      | 38.8            | 47.9 | 64.3 | 56.1 | 8.2  | 8.0  | 65.1       | 68.7 | 72.0 | 63.9 | 9.3  | 7.5 |
| +3 distractors      | 40.9            | 47.0 | 63.1 | 55.1 | 9.7  | 8.2  | 64.5       | 68.0 | 68.4 | 62.1 | 11.7 | 6.9 |
| +4 distractors      | 41.9            | 45.3 | 62.5 | 52.5 | 9.5  | 6.6  | 63.5       | 68.0 | 72.2 | 62.5 | 10.5 | 8.7 |
| +5 distractors      | 39.5            | 42.6 | 62.3 | 54.4 | 10.0 | 6.9  | 64.6       | 69.1 | 70.2 | 60.8 | 11.4 | 7.3 |
| <i>2 gold pages</i> |                 |      |      |      |      |      |            |      |      |      |      |     |
| Oracle              | 28.9            | 33.3 | 37.8 | 34.6 | —    | —    | 62.2       | 63.4 | 70.4 | 61.5 | —    | —   |
| +1 distractor       | 25.7            | 34.0 | 38.6 | 34.4 | 7.8  | 7.5  | 58.7       | 63.0 | 66.2 | 57.9 | 9.6  | 4.1 |
| +2 distractors      | 25.3            | 33.6 | 35.7 | 32.4 | 9.6  | 6.0  | 58.9       | 62.2 | 66.4 | 57.4 | 10.9 | 4.4 |
| +3 distractors      | 21.6            | 28.2 | 33.6 | 33.2 | 11.4 | 6.0  | 58.1       | 63.3 | 70.8 | 57.8 | 9.7  | 6.0 |
| +4 distractors      | 24.6            | 30.7 | 43.7 | 31.3 | 11.8 | 11.8 | 57.8       | 63.2 | 68.3 | 57.1 | 13.3 | 6.1 |

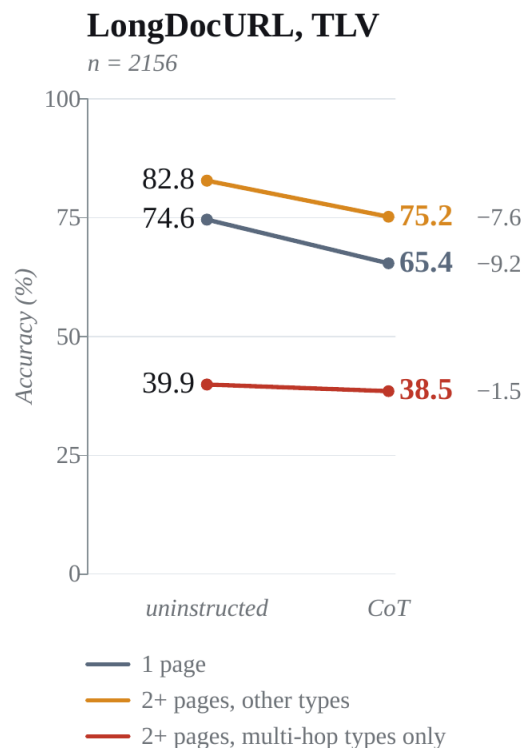
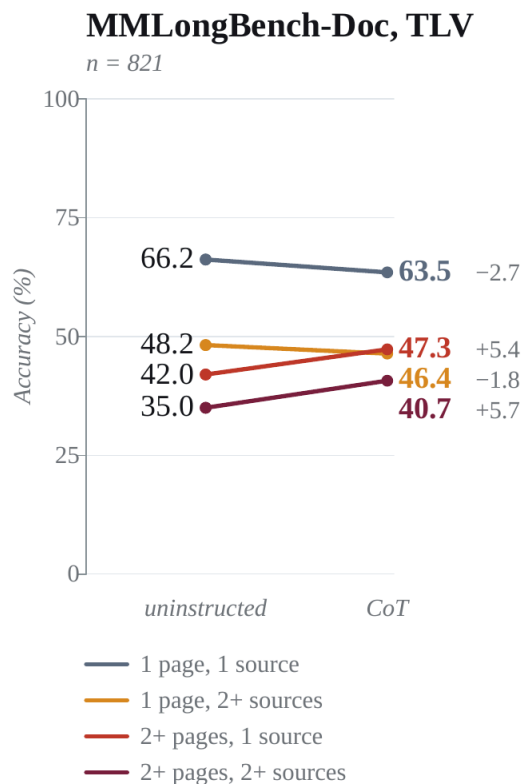
A complete gold set, then the same set plus one to five distractor pages. Tested range only: context stays at six pages or fewer

## READING IT

- **Aggregate accuracy barely moves.** Single-page questions stay broadly stable, and multi-page questions decline gradually. LongDocURL degrades even less under comparable distractor counts.
- **But stability is not harmlessness.** The verdict transitions show added pages both breaking and repairing answers. Positive and negative effects partly cancel at the aggregate level.
- **A non-gold page is not always irrelevant.** Sometimes it adds a partial instance that makes the model overcount; sometimes it supplies the context that disambiguates the gold pages and repairs the answer.
- **So lean toward coverage** when the later context can absorb it, and control how the retrieved evidence is presented rather than minimising how much of it there is.

# Reasoning Across Pages Remains Hard

Every correct page is supplied, in the richest representation. **Cross-page questions are still much harder**, and asking for step-by-step reasoning helps only some of them.



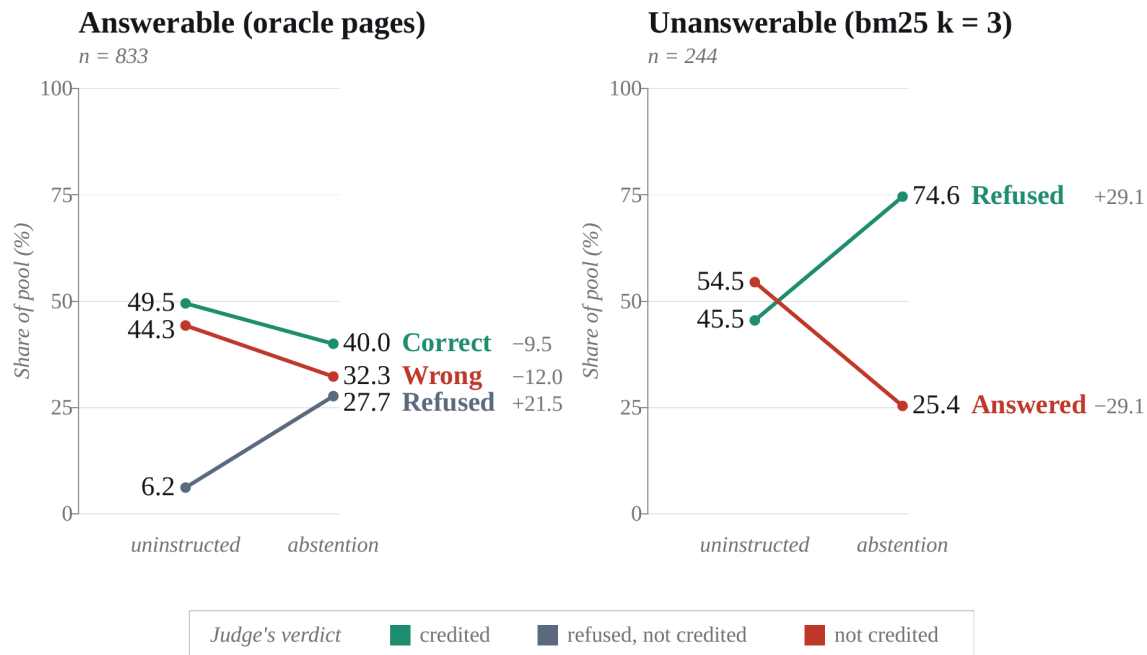
## READING IT

- **The deficit is real.** On MMLongBench-Doc, one-source questions fall from 66.2% on a single page to 42.0% once the evidence spans pages, and to 35.0% when several sources must be combined as well.
- **Chain-of-thought helps precisely there.** It slightly hurts single-page questions (66.2 to 63.5), and lifts cross-page ones (42.0 to **47.3**, and 35.0 to **40.7**). The pooled integration deficit narrows from 24.6 to 18.2 points.
- **Most of the gain is structured evidence.** Tables improve by **10.9** points; charts, text and layout by two to three; figures fall slightly.
- **And it does not generalise.** On LongDocURL the same prompt reduces accuracy almost everywhere. More reasoning helps when it matches the task, not because there is more of it.

Effect of chain-of-thought across evidence structures, TLV rung, oracle evidence pages

# Abstention Is Not Well Calibrated

The last reasoning decision is whether the evidence supports an answer at all. An abstention instruction makes the model more cautious, **but not more discerning**.



Effect of targeted abstention prompting at the TLV rung

## READING IT

- **It works where it should.** On unanswerable questions, refusal rises from 45.5% to **74.6%**, cutting unsupported answers substantially.
- **And it fires where it should not.** On answerable questions, refusal rises from 6.2% to **27.7%**, and correct answers fall from 49.5% to 40.0%.
- **Difficulty is being read as insufficiency.** Under the instruction, refusal reaches 35.5% for figure questions and 30.3% for tables, against 24.1% for text — with oracle pages supplied, so the evidence is definitely there.
- **The prompt moves willingness, not judgement.** It shifts the operating point rather than separating sufficient evidence from insufficient.

# A Simple Adaptive Method

The survey and the experiments both point at design principles. I put them together in a **deliberately simple system**, to see whether they hold up at the level of a whole pipeline rather than one controlled condition.

## REPRESENTATION

### Keep text and images together

- ▶ Every page read at the full TLV representation, with a strong parser and high-resolution rendering, since the two channels cover for each other's failures.

## SELECTION

### Gather broadly, present selectively

- ▶ Missing evidence is unrecoverable and moderate distraction is not, so read widely in retrieval-ranked order — but carry most of it forward as compact records rather than as full pages.

## REASONING

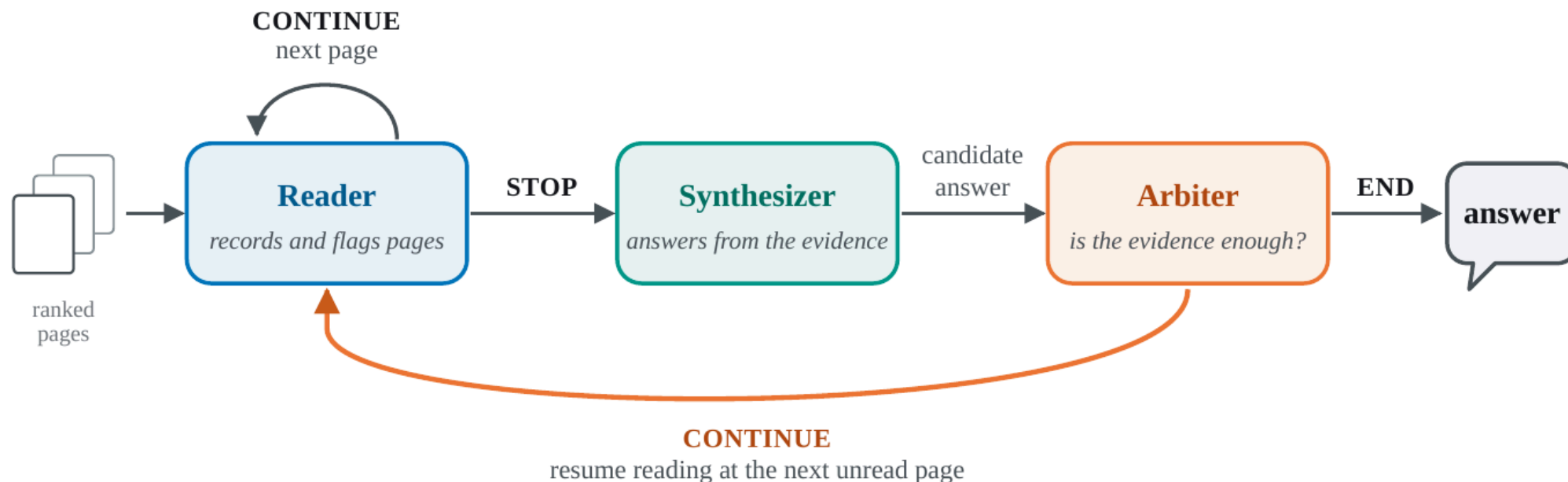
### Check whether the answer is supported

- ▶ Because the reasoner cannot judge sufficiency on its own, a separate step decides whether to accept the answer or keep reading.

No planner, no tool set, no query reformulation. **The point is to validate the principles, not to introduce a complex architecture.**

# Reader, Synthesizer, Arbiter

One underlying model in **three roles**, differentiated only by the instruction and the input each receives.



## reader

Works through the pages in retrieval-ranked order, recording answer-relevant evidence as it goes, and proposes stopping when it thinks it has enough.

## synthesizer

Receives the **most important pages in full**, plus the short records from everything else read, and proposes an answer from that.

## arbiter

Accepts the answer, or sends the reader back for more. It cannot write an answer itself, which makes **stopping too early recoverable** rather than final.

The loop is what makes the amount of reading **adapt to the question**: an easy one stops early, a hard one keeps going.

# System-Level Validation, and Its Cost

On MMLongBench-Doc the method reaches **54.5%** under the official scorer, above the gold-page baseline using the same model family.

| Method                                                   | OCR | Trained        | Evidence source |             |             |             |             | Evidence pages |             |             | ACC         |
|----------------------------------------------------------|-----|----------------|-----------------|-------------|-------------|-------------|-------------|----------------|-------------|-------------|-------------|
|                                                          |     |                | TXT             | LAY         | CHA         | TAB         | FIG         | SIN            | MUL         | UNA         |             |
| Official protocol: answer extraction + rule-based scorer |     |                |                 |             |             |             |             |                |             |             |             |
| Gold pages + VLM (TLV)                                   | ✓   | ×              | 48.1            | 49.6        | 39.4        | 50.5        | 42.5        | 57.7           | 37.4        | 52.5        | 49.5        |
| <i>Qwen2.5-VL-7B</i>                                     |     |                |                 |             |             |             |             |                |             |             |             |
| MDocAgent <sup>†</sup>                                   | ✓   | ×              | 43.1            | 31.2        | 32.6        | 38.7        | 37.9        | 53.5           | 23.8        | 28.3        | 38.5        |
| MoLoRAG+                                                 | ×   | ✓ <sup>a</sup> | 42.7            | 38.8        | 33.3        | 38.5        | 40.7        | 52.9           | 27.6        | 35.9        | 41.0        |
| MM-Doc-R1 <sup>†</sup>                                   | ✓   | ✓              | 46.2            | 38.1        | 40.8        | 52.8        | 35.9        | 56.0           | 31.2        | 68.2        | 49.7        |
| <i>Qwen3-VL-8B</i>                                       |     |                |                 |             |             |             |             |                |             |             |             |
| Whole document                                           | ×   | ×              | —               | —           | —           | —           | —           | 46.9           | 35.3        | 25.8        | 38.5        |
| MARDoc                                                   | ✓   | ×              | —               | —           | —           | —           | —           | —              | —           | —           | 52.7        |
| DocTrace                                                 | ✓   | ✓              | —               | —           | —           | —           | —           | 53.2           | <b>41.1</b> | 70.5        | 52.9        |
| Ours (4-bit)                                             | ✓   | ×              | <b>48.0</b>     | <b>41.9</b> | <b>43.7</b> | <b>59.9</b> | <b>41.8</b> | <b>58.9</b>    | 37.8        | <b>71.9</b> | <b>54.5</b> |

MMLongBench-Doc accuracy, official protocol. Published systems on open 7-8B backbones, values taken from the cited papers

## READING IT

- Above the oracle baseline. **54.5%** against **49.5%** for the same model reading only the annotated gold pages, which suggests useful context comes from **beyond** the pages the benchmark labels as evidence.
- And above the published comparisons on the same backbone, the strongest being 52.9%.
- The gain is not uniform. Multi-page questions reach 37.8%, still below DocTrace's 41.1%. Cross-page integration remains the hard part, exactly as the attribution study predicted.

8.2 GiB

mean peak VRAM

13.9

agent turns per question

9.3 min

median latency per question

16.6 min

mean, almost all decoding

# Conclusion

Long-document understanding is, above all, a matter of **managing evidence well**.

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## Preserve

What survives the conversion of a page into text, pixels, and structure.

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## ► Select

Which of it actually reaches the model, out of hundreds of pages.

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## ► Reason

Whether the model combines it, and knows when it is not enough.

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### 1 The survey

maps how systems preserve, find, and use information across pages.

### 2 The empirical study

shows where the failures arise, and that each one needs a different fix.

### 3 The simple method

shows those lessons can be turned back into a working system.

**Thank you — happy to take questions.**